Affect as Information about Users' Attitudes to Conversational Agents

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Abstract

This paper presents a novel method for automatic evaluation of conversational agents. In the method, information about users' attitudes and sentiments to conversational agents and their performance are achieved by analyzing their general emotional engagement in the conversation and specific affective states, and interpreting them using psychological reasoning of Affect-as-Information. In the evaluation experiment the users' attitudes to two Japanese-speaking conversational agents were checked simultaneously in a survey and using a system constructed on the proposed method. The results returned by the system revealed similar tendencies as the survey. Therefore the method is applicable as a mean of evaluation for Japanese-speaking conversational agents.

1. Introduction

Technological development focused on enhancing and facilitating human lives led to creating a need for intelligent environment meeting all human needs, like MIT's House_*n* [1]. Explorations in the field of Ambient Intelligence [2] brought to light a new dimension of communication, where a machine becomes an interlocutor for a user, Human-Computer Interaction (HCI) [3]. What was carried whit it, was a rush in development of intelligent talking agents in car navigation systems [4] or even furniture [5]. Their functional implementation into our lives has already become a current process. A need for environment not only intelligent, but also humanized grows rapidly [6].

Along with it people found themselves in a need for developing a quick, automatic evaluation methods for such agents. The usual methods used are based on asking the opinions of users-testers about their satisfaction during interaction with an agent, the will for continuing the conversation, the naturalness of agent's utterance producing, etc.

However, since decision making and therefore expressing opinions in humans strongly depend on features like emotional states, imagination and experiences [7, 8], usual surveys are always stigmatized with potential lack of objectivity. To cope with this problem, we propose a novel method of estimating the user's attitude and sentiment towards a Japanese-speaking agent, based on a "affect as information" [9] reasoning derived from affect analysis of the user's utterances.

2. Affect as information

The notion of "affect as information" was introduced in 1983 by Schwarz and Clore [9] and is widely studied in the field of psychology and social psychology. Schwarz and Clore claimed that people use affect just as any other criterion, by applying the informational value of their affective reactions to form their judgments, attitudes and opinions.

As we assumed, applying this thought to evaluation of conversational agents should indicate similar tendencies in the results of attitudes classification and the results acquired through a survey. If this approach appeared to be correct, it would be as step towards practical realization of the idea of affective human factors design [10], with the information about a product (agent) being derived from information about dynamic changes of the user's affective states during usage. Therefore, in the process of product design, the affective factors become not only as noteworthy as usability [10], but in some cases, the affect itself can provide valuable information about the usability.

3. Affect derived sentiment

3.1. Sentiment analysis for agent evaluation

Sentiment analysis is a sub-field of information extraction that only recently has gained interest of

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scientists [11]. The general idea of sentiment analysis, which is to gather and classify attitudes (into positive or negative) about particular topics or entities, is important for marketing research, monitoring of chat-rooms' content for security reasons [12], or customer feedback on particular products. It would be desirable to acquire objective information about agents' performance before putting them on the market, since a failure might bring a great loss of funds and human effort. Tests, where people are hired to verify the performance of market-destined agents, are burdened with a load of effort and funds. Moreover, paying people a lot of money for the evaluation undermines the test objectivity. Although there is no other way of performing a test, then making a human talk to an agent, there is a method of gathering more objective information for the evaluation then a typical survey.

In our assumption, gathering information about the tester's sentiment towards the product during the test will provide the objective information. We gather it by analyzing the users' affective states during conversations and, use the "affect as information" reasoning to determine the user's attitude towards its interlocutor – the agent.

3.2. Affect analysis for sentiment estimation

Affect analysis is a sub-field of information extraction. Its goal is to estimate human emotional states in communication. Popular methods for analyzing affect include analyzing emotions from facial expressions or voice [13]. However, since emotions are strongly context dependent [14], most of the semantic content of expressing emotions is ignored in such researches. Therefore we decided to use a method to analyze the affect of a textual representation of an utterance.

There are several researches on affect analysis [15]. However there have been only few approaches to apply the affect analysis to gather information about sentiment and attitudes [16] and no significant work was done on applying such approach to evaluation of conversational agents talking in Japanese.

4. ML-Ask for affect analysis

As the affect analysis system we used Ptaszynski's at al. Emotive Elements/Emotive Expressions Analysis System (ML-Ask) [15]. The system is designed to recognize affect in textual input in the Japanese. The process of analysis in the system is performed in two general steps: 1. Determining whether an utterance is emotive or not (general emotiveness); 2. In emotive utterances finding the specific types of emotions conveyed.

The method of analyzing the affect is based on Ptaszynski's [17] idea of binary classification of realizations of emotions in language into:

1. *Emotive elements.* Informing that emotions have been conveyed, but not expressing specified feelings or expressing different ones, depending on the context. Examples are: *sugee* (great!), *waku-waku* (heart pounding), *-yagaru* (fu**ing do sth);
2. *Emotive expressions*. Parts of speech, which, in emotive sentences, describe emotional states. Examples are: *aijou* (love), *kanashimu* (feel sad), *ureshii* (happy).

In a textual input provided by the user three described above features are determined gradually by referring first to the databases of emotive elements and then the emotive expressions. The emotive elements databases were gathered from different research [18, 19, 20, 21] and divided into interjections, mimetics, endearments, vulgarities and representations of non-verbal emotive elements, like exclamation marks or ellipsis. The databases of emotive expressions were created on Nakamura's [22] collection.

4.1 ML-Ask's output as information

We use ML-Ask to analyze utterances of a user talking to a conversational agent. The results of analysis are then viewed as follows. First, if many[1] of the user's utterances were determined as emotive, we assume the user was emotionally involved in the conversation. In the Japanese, emotional involvement in a conversation suggests a tendency to easier familiarizing with the interlocutor, and therefore, considering the machine as more human by loosing the sense of recognizing a mere machine in the agent. Second, analysis of specified emotion types conveyed by the user in the whole conversation provides information on what were the user's feelings towards the machine interlocutor during the conversation. If the feelings were positive, or changing from negative to positive while talking, the attitude, and therefore the general sentiment towards the agent is considered as positive. If the emotions were negative or changing from positive to negative, the sentiment is classified as negative.

Both types of information acquired provide a wide overview on the user's sentiment about the agent and it is desirable for the both types of information to harmonize rather then show dissonance. Such analysis, if accurate, realizes the first step in affective human factors design, which is to understand the user's affective needs [10].

[1] We do not precise the value of "many". Instead of that we compare the results for two different conversational agents.

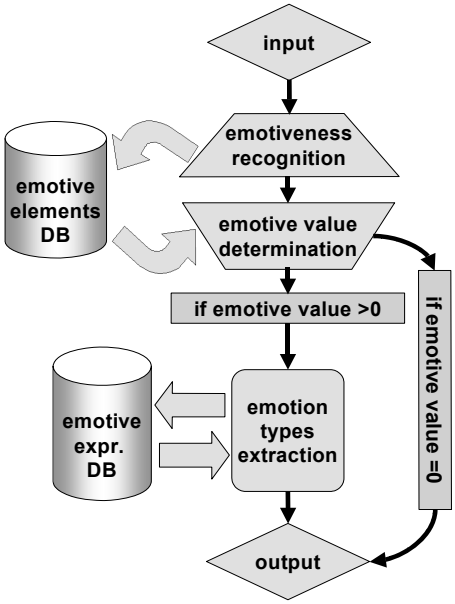


Fig. 1. Flow chart of ML-Ask.

5. Applying 2-dimensional model of affect

The idea of 2-dimensional model of affect was first proposed by Schlosberg [23] and developed further by Russell [24]. Its main assumption is that all emotions can be described in a space of two-dimensions: the emotions' polarity (positive / negative) and activation (activated / deactivated). An example of positive-activated emotion would be an "excitement"; a positive-deactivated emotion is, for example, a "relief"; negative-activated and negative deactivated emotions would be "anger" and "gloom" respectively.

5.1. Classification of emotions

Nakamura [22], after a thorough study on emotions in the Japanese, proposes a classification of emotions into 10 types - said to be the most appropriate for the language. That is: [joy/delight], [anger], [sorrow/sadness], [fear], [shame/shyness/bashfulness], [liking/fondness], [dislike/detestation], [excitement], [relief] and [surprise/amazement].

5.2 Nakamura's emotions on Russel's scale

We grouped the emotion types distinguished by Nakamura on the Russell's model (see Fig. 2). The types fitting in two groups, like *kou* / *takaburi*, which contains expressions of positive "excitement" as well as negative "irritation", were put in both possible quarters. However none of the types was placed in more than two quarters.

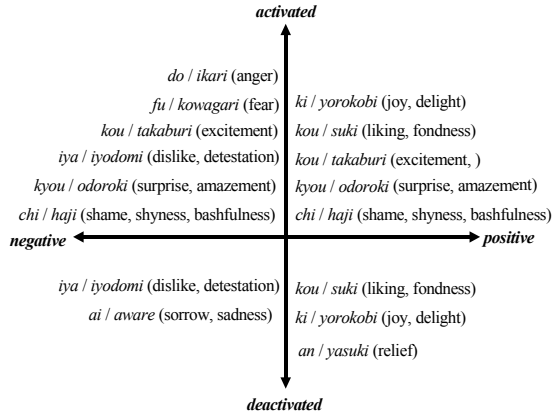


Fig. 2. Grouping Nakamura's classification of emotions on Russell's two-dimensional space.

6. Evaluation experiment

To test the method we performed an evaluation experiment of the two conversational agents. There were 13 participants of the experiment, 11 males and 2 females; all of them were university undergraduate students. The users were asked to perform a 10-turn dialogue with Modalin, and then with Pundalin. No topic restrictions were made, so that the talk could be as free and human-like as possible. The agents were first evaluated during the conversation using the proposed method. After the conversation the users were asked to fulfill a questionnaire concerning their sentiment about the agents and each system's performance. The results acquired by the method and provided by the users for the two agents were compared looking for similarities in sentiment classification.

6.1 Two agents – a short description

6.1.1. Modalin. A non-task oriented keyword-based conversational agent, which uses modality to enhance Web-based propositions for dialogue. The agent was developed by Higuchi et al. [25].

6.1.2. Pundalin. A non-task oriented conversational agent created on the base of Modalin combined with Dybala's Pun generating system PUNDA [26]. Pundalin therefore is a humor-equipped conversational agent using puns to enhance the communication with a user.

6.2. Questions we asked users and their representations in sentiment analysis

6.2.1. User's evaluation. The questions asked were: A) Do you want to continue the dialogue?; B) Was the agent's talk grammatically natural?; C) Was the agent's talk semantically natural?; D) Was the agent's vocabulary rich?; E) Did you get an impression that the agent possesses any knowledge?; F) Did you get an impression that the agent was human-like?; G) Do you think the agent tried to make the dialogue more funny and interesting? and H) Did you find agent's talk interesting and funny?. The answers for the questions were given in 5-point scale with some explanations added. Each user filled two such questionnaires, one for each agent. The final, summarizing question was "Which agent do you think was better?"

6.2.2. Representations in sentiment analysis. We made the following assumptions on how the questions we asked users directly were represented by the results provided by the analysis. The questions A)–H) generally illustrate how high did the users evaluate agents' talking abilities (B-D), much users could familiarize with agents (A, E-F) and engage their feelings in the conversation (A, G-H). Therefore the particular questions represent the first type information as well as the information from the activated/ deactivated dimension provided from the analysis (for details see section 5). The general summarizing question represented the polarity dimension of negative / positive attitudes acquired from the second type of information provided by the analysis.

7. Results

The results of the evaluation are showed below. First the results of the questionnaire are shown and after that the results of the analysis are summarized and compared to the user's direct opinion.

7.1 User's evaluation

11 out of 13 users (84.6%) evaluated Pundalin (humor-equipped agent) as better than Modalin (see Fig.3). Pundalin received higher scores also in detailed questions (see Table 1. and Fig. 4.).

Although not in all categories the differences between Modalin and Pundalin were that significant, overall results for both agents clearly showed, the performance of which was estimated as a more human-like and easy to familiarize.

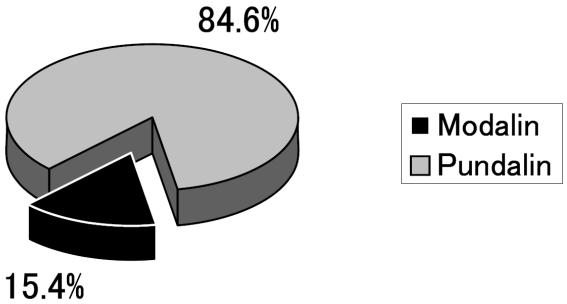


Fig. 3. User's evaluation-results for the question "Which agent do you think was better?"

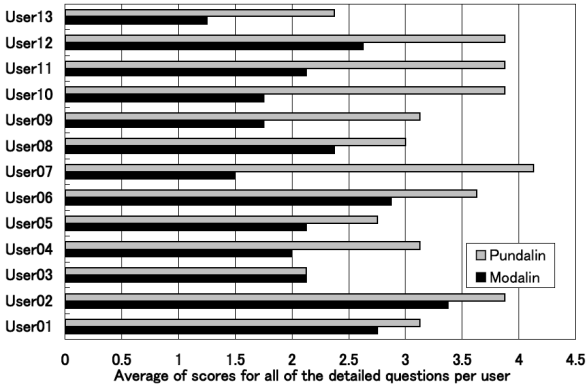


Fig. 4. User's evaluation - results for Modalin and Pundalin for detailed questions per user (see 5.2.1.). Answers were given in a 5-point scale.

Table 1 User's evaluation - results for Modalin and Pundalin for detailed questions per question.

Questions	A	B	C	D	E	F	G	H
Modalin	2.62	2.15	1.85	2.08	2.15	2.38	1.92	2.46
Pundalin	3.38	2.92	2.69	3.00	2.85	3.31	4.15	4.08

7.2. Results of sentiment analysis

Evaluation based on sentiment analysis of the users' utterances showed tendencies similar to the. The users were more emotionally involved in the conversations with Pundalin, which corresponds to the direct opinions about the agent – that it was more human like, its utterances were more correct semantically, grammatically, etc. (see 6.2.1. and 7.1., Fig. 5.) and therefore easier to familiarize.

The analysis of specified emotion types conveyed by the users in conversations provided the information clearly showing the users' feelings and attitudes towards the both agents. The users' general attitudes to Pundalin were in 78% positive (see Fig. 7.) whereas to Modalin the attitudes of the users were mostly (75%) negative (see Fig. 6.).

Therefore we can say that the general sentiment of a user towards an agent was better for Pundalin than for Modalin.

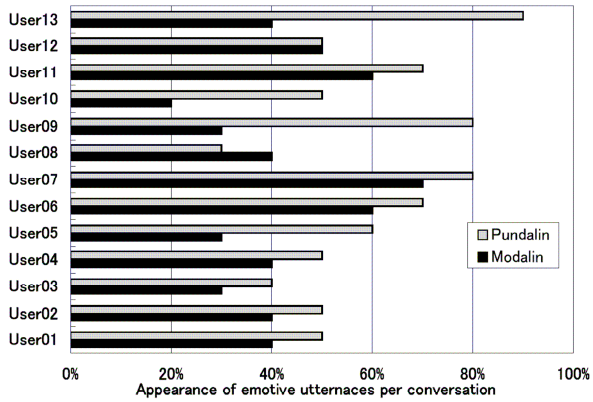


Fig. 5. Percentage of average appearance of emotively engaged utterances for all 13 users in conversations with both agents.

8. Conclusions

In this paper we presented a novel method of evaluation for conversational agents. The method is based on analyzing affect conveyed by a user in a conversation with an agent. Borrowing the notion of "affect as information" [9], the results of affect analysis performed by a system created by Ptaszynski, et al. [15] provide us information about the user's emotional involvement in a conversation, loosing a psychological distance, and easiness in familiarizing with the machine. This corresponds to the direct questioning the user about the agent's performance. Next, analysis of specified emotion types conveyed by the user in the whole conversation and their classification by applying the two-dimensional model of emotions [24] provides us information on what was the polarity of the user's attitudes towards the machine interlocutor during the conversation.

In the evaluation experiment performed on two conversational agents there have been seen similar tendencies in the results acquired by the method and the results of the questionnaire. The results are not in 1:1 ratio, which is almost impossible to achieve, and the attention should rather be focused on the similarities in tendencies in general comparison of the two agents. The proposed method, comparing to the traditional user-oriented surveys, is non-invasive and provide objective information on user's sentiment about machine interlocutor. Since an approximate time of processing one utterance is below 0.15s, the method can be used in real time, and provide actual information on changes in the user's attitude towards the machine. This feature does not only provide a fast and up-to-date information on the user's sentiment, but also, appropriately utilized, can provide hints for the agent about the potential undesirable changes in the user's attitudes and the need for appropriate counteractions, during an everyday use.

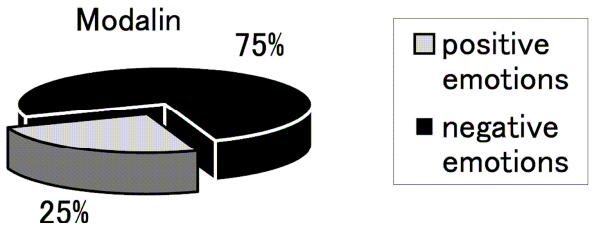


Fig. 6. The total relation of emotions positive to negative conveyed in the utterances of users with Modalin.

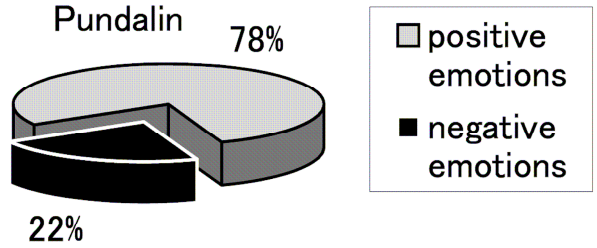


Fig. 7. The total relation of emotions positive to negative conveyed in the utterances of users with Pundalin.

By applying the proposed method in evaluation of conversational agents, the evaluating information is acquired in the process of testers conversing with an agent. Therefore as an evaluative mean, the method saves time, effort and funds spend each time on preparing and performing laborious surveys.

It is desirable for the proposed method to be accepted widely in field as a full equivalent or at least a strong supportive mean to objectivize the results of traditional questionnaires.

9. Discussion and future work

The method, although proved to be effective, has still some lacks, desirable to be filled in the near future. The imperfections of the sub-systems used in the method influence its accuracy. The slight lacks in emotion types extraction procedure in ML-Ask limit the information about affective states conveyed by users in a conversation. However, it is predictable that applying the two-dimensional model of emotions (see section 5) into assigning emotional affiliations of emotive elements will disambiguate the emotional affiliations of emotive elements improving the performance of ML-Ask.

The system is designed for the Japanese language, however constructing a different language version of ML-Ask is possible after gathering adequate databases in other languages, especially the ones with similar morphology, like Korean.

The notion of "affect as information", although with a firm scientific background in psychology and social psychology [27, 28], is not a common notion in the fields we referred to in this paper - development of agents, evaluation methods or affect analysis. The mapping of questions on the results of affect analysis was, although supported with strong theory, still rather commonsensical and intuitive, therefore we will tend to precise the mapping to be more evident. However, in this experiment we tried to prove that affective states do influence judgments and attitudes and, properly analyzed, reveal similar tendencies as usual evaluation surveys, providing valuable and important information in evaluation - a significant part of the product design process. Coordinating the appropriate items to evaluate automatically with the questions asked directly to the user is based on psychological reasoning, and therefore reaches deeper and beyond the simple numbers usually put it in terms of so familiar notions of precision or recall. However, the rapid development in all fields of science, as well as in commercial areas, makes researchers from different scientific fields join efforts, which - as we shown in this paper - can be successful.

Acknowledgments

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